

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.21_K-1)

Metric

$$ds^2 = \left(\frac{2m+r}{r} \right) du \otimes du - du \otimes dr - dr \otimes du + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dx \otimes dx + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{m}{r} - \frac{1}{2} \right) du + (1) dr$$

$$m = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(-\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

$$\bar{m} = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) \, du + \left(-\frac{m}{r} - \frac{1}{2} \right) \, dr$$

$$m = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(-\frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \right) dy$$

$$\overline{m} = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(\frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{m}{r} - \frac{1}{2} \partial_r$$

$$\delta = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x - \frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \partial_y$$

$$\bar{\delta} = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x + \frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{m}{r^2} - \frac{1}{2r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\beta = \frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{m}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{m}{r^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "